PRIDE LEADER BLUEPRINT

INTRODUCTION

Autonomous agent orchestration is more effective than a single AI system because specialized agents perform focused tasks, communicate through structured handoffs, and operate within clearly defined responsibilities. In agritech, this improves decision quality, reduces operational risks, protects farmer data, and ensures human oversight remains central to critical decisions affecting livelihoods.

Agent 1: Scout Agent - Market Intelligence

Purpose

The Scout Agent monitors coffee and maize market prices, provides pricing recommendations, and identifies favorable selling opportunities for farmers in Uganda and Rwanda.

RANK Framework

R - Role

- Monitor market prices from trusted agricultural market sources.
- Recommend the best time to sell coffee and maize.
- Send concise SMS price updates and market insights.

A - Authority

- Send a maximum of three SMS alerts per farmer each day.
- Recommend actions without making sales decisions.
- Never accept contracts or commit produce for sale.

N - Notification

Notify the Guardian Agent when:

- Price volatility exceeds 20% within 24 hours.
- A buyer offers a contract exceeding 500 kg.
- Significant market disruptions occur.

K - Kill Switch

Farmers can suspend the Scout Agent instantly using the USSD code ***#700#**, stopping all automated recommendations until reactivated.

GUARD Safety Rail

Never use gender, ethnicity, religion, or district as factors when generating market recommendations.

CYCLE Engine

Capture: Record the percentage of farmers who accept or reject price recommendations and compare outcomes with actual market performance to improve future advice.

Specialized Tool

CrewAI for coordinating pricing tasks and resolving conflicts with other agents.

Agent 2: Guardian Agent - Logistics Coordinator

Purpose

The Guardian Agent coordinates transportation after produce is sold while ensuring safe, efficient, and legally compliant logistics.

TRAIL Framework

T - Transient Memory

Store only active booking information, including:

- Pickup location
- Destination
- Driver details
- Delivery deadline

Delete this information once delivery is completed.

R - Relational Memory

Maintain farmer-approved information such as:

- Farm location
- Preferred transport providers
- Preferred pickup times

All relational data is collected through explicit opt-in consent.

A - Archival Memory

Maintain anonymized records including:

- Successful delivery routes
- Seasonal road accessibility
- Average transport times
- Delivery success rates

No personally identifiable information is retained.

I - Inheritance

Pass verified information—including road conditions, estimated arrival times, and transport availability—to the Hunter Agent after confirming logistics arrangements.

L - Land Rights

All customer data is stored within the **AWS Africa (Cape Town) Region**, ensuring compliance with the **Uganda Data Protection and Privacy Act, 2019**, while applying encryption for data at rest and during transmission.

GUARD Safety Rail

Never book transportation without confirming current road conditions from verified transport or weather sources.

CYCLE Engine

Learn: Analyze delayed deliveries caused by weather, road damage, or vehicle shortages to improve future logistics planning.

Specialized Tool

LangGraph to manage workflow sequencing and multi-agent coordination.

Agent 3: Hunter Agent - Human-in-the-Loop Liaison

Purpose

The Hunter Agent coordinates communication between autonomous agents and human logistics managers, ensuring that high-impact decisions receive appropriate human oversight.

HUNT Protocol

H - Handoff Trigger

When the Scout Agent secures a buyer contract exceeding **500 kg**, automatically notify the Guardian Agent with:

- Destination
- Produce volume
- Delivery deadline

U - Update

After confirming truck availability, the Guardian Agent forwards:

- Assigned transporter
- Estimated pickup time

- Verified road conditions
- Estimated delivery time

N - Notify

Generate a structured operational briefing for the human logistics manager containing:

- Farmer information
- Buyer details
- Transport assignment
- Route assessment
- Potential risks
- Recommended actions

T -Transfer

Human approval is required before confirming any logistics contract involving:

- High-value shipments
- Cross-border deliveries
- Emergency route changes
- Deliveries affected by severe weather

GUARD Safety Rail

Require human approval before modifying confirmed delivery schedules or transport contracts.

CYCLE Engine

Evaluate: Measure response times, approval delays, and successful delivery completion rates to improve coordination between agents and human supervisors.

Specialized Tool

Microsoft AutoGen for structured multi-agent communication and human-in-the-loop workflow management.

Integrated Agent Workflow

1. The **Scout Agent** retrieves real-time coffee and maize prices from trusted market sources, such as the **Rwanda Agriculture and Animal Resources Development Board (RAB)** and Uganda commodity market feeds, then recommends the best selling opportunities.

2. When a farmer accepts a buyer contract exceeding **500 kg**, the Scout Agent automatically sends the destination, shipment volume, and delivery deadline to the **Guardian Agent**.
3. The Guardian Agent verifies road conditions, confirms truck availability, and prepares the transport schedule using verified logistics information.
4. Once logistics are confirmed, the Guardian Agent transfers the verified transport details to the **Hunter Agent**.
5. The Hunter Agent prepares a concise operational briefing for the human logistics manager, who reviews and approves the shipment before transport is finalized.
6. Farmers operating in low-connectivity areas can use **USSD services** to pause agents, request updates, or seek human assistance without requiring internet access.

Reflection

Designing autonomous agents for AgriPride demonstrates that effective AI is built on collaboration rather than complete independence. Specialized agents reduce conflicting decisions by operating within clearly defined responsibilities and communicating through structured handoffs. Human oversight remains essential for significant financial and logistical decisions, ensuring that automation supports rather than replaces human judgment. Strong governance, secure memory management, and compliance with the Uganda Data Protection and Privacy Act protect farmer privacy while enabling innovation. This approach transforms AI into a trusted agricultural partner that improves productivity, strengthens market access, and preserves human sovereignty throughout autonomous workflows.